

WONG, Hok Fong

Intern, Summer@EPFL; BSc in Computer Science, CUHK

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ACADEMIC INTERESTS

Machine Learning and Theory; Optimization; Design and Analysis of Algorithms; Theoretical Computer Science; Atmospheric Science

EDUCATION

The Chinese University of Hong Kong, Hong Kong [Cumulative GPA: [3.803 / 4.0](#) | Major GPA: [3.916 / 4.0](#)] *Aug 2022 - Jun 2026*
Major: **B.Sc. in Computer Science** with **First-Class Honors** (Focus: Algorithms and Complexity) [Academic Transcript](#)
Minors: **Mathematics; Earth and Environmental Sciences** (Focus: Atmospheric Science) (151 Units = ~302 ECTS [\[Ref\]](#))
Engineering Enhancement Program: **Engineering Leadership, Innovation, Technology and Entrepreneurship Stream (ELITE Stream)**

PUBLICATION

H. F. Wong, H.-T. Wai, C.-Y. Yau, "Large Stepsizes Federated Learning on Logistic Regression with Linearly Separable Data: The Case of Heterogeneous Devices," *The 65th IEEE Conference on Decision and Control* ([accepted](#)), 2026. ([Theory Paper](#))

WORK EXPERIENCES

(Ongoing) Junior Research Assistant – Dept. of Earth and Environmental Sciences, CUHK *Sep 2026 - Aug 2027*
- **Atmospheric Chemistry Modeling and Climate Group**. Actual responsibilities [TBD](#). Tentatively on AI/ML for atmospheric chemistry, through *developing a global anthropogenic fugitive dust emission inventory with multi-source data* with the aim of publishing a paper as main author.
- **Official teaching assistant of the undergraduate/research postgraduate course EESC4550/5550** – Practical Atmospheric Modeling (2026 Fall), involved in course content development (Linux HPC Computing, AI/ML Toolbox for Atmospheric Chemistry), design of assignments and lab exercises (Modeling Stratospheric Ozone Dynamics, GEOS-Chem Tutorials). (Available upon request.)
- (Upcoming) Teaching assistant of the taught postgraduate course AISC5010 – Basic Programming for AI (2027 Spring).
- **Supervisor: Prof. Shixian ZHAI**, Department of Earth and Environmental Sciences, CUHK.

Summer Research Fellow – Summer@EPFL 2026, Switzerland *Jun 2026 - Aug 2026*
- Selected from 7,000+ global applicants ([1.2% acceptance rate](#)), with the focus on *optimization for machine learning (theoretical analysis)*.
- On *high-probability guarantees of SGD with linear warm-up* under (H_0, H_1) -smoothness and aiming condition, targeting ICML 2027; the result directly addresses two major weaknesses of the state-of-the-art analysis: impractical dependence on f^* and the interpolation requirement. [\[Slides\]](#)
- **Supervisor: Prof. Martin JAGGI and Dr. El Mahdi CHAYTI**, Machine Learning and Optimization Laboratory (MLO), EPFL.

Research Student – CUHK Undergraduate Summer Research Internship (2024, 2025) *Jun 2024 - May 2026*
- Conducted research in modern optimization methods tailored for *large-scale machine learning*, focusing on stochastic methods for nonconvex problems.
- Analyzed theoretical properties such as *convergence guarantees, and complexity bounds* under smoothness and/or bounded gradient assumptions.
- Awarded *research grants* from Faculty of Engineering, with a total amount of **HKD \$28,000**.
- **Supervisor: Prof. Hoi-To WAI**, Department of System Engineering and Engineering Management, CUHK.

Curriculum Designer – CUHK & Jockey Club "AI for the Future" Project *Mar 2023 - Dec 2024*
- Spearheaded the *creation of engaging courseware* introducing the knowledge of Artificial Intelligence (AI) to thousands of secondary school students.
- Collaborated with professors and post-doctoral researchers to design curriculum content aligned with the *first AI syllabus launched for junior secondary education in Hong Kong*. [\[Website\]](#)
- Received stipends with a total amount of **HKD \$12,000** for the technical quality and the education impact of the work conducted.
- **Supervisor: Prof. Mei Ling Helen MENG, Dr. Xing XING**, Stanley Ho Big Data Decision Analytics Research Centre, CUHK.

SELECTED PROJECTS

A more detailed list of projects, essays, and codes are available: [\[Link\]](#)

Theoretical Analysis of Large-Stepsize Gradient Descent on Multinomial Logistic Regression *Fall 2025, Spring 2026*
- *Graduation thesis challenging conventional smoothness-based convergence limits for large-stepsize Gradient Descent*. [\[PDF\]](#)
- Rigorous theoretical analysis of the *optimization dynamics of multinomial logistic regression with separable data*, delivering its *first non-asymptotic convergence bound*. Precisely quantified that *loss and phase transition time scale only logarithmically with the number of classes*.
- Extended analysis to *FedAvg with large stepsizes and heterogeneous local updates* where heterogeneity vanished asymptotically. Also characterized convergence on *non-separable data*: adaptive stepsizes guarantee monotonic $O(1/t)$ convergence, constant-stepsize cycles exist for $K \geq 3$ classes.

On Quantization-Aware Training Algorithms and Stronger Convergence Guarantees of ASkewSGD *Summer 2024, 2025*
- *Research project* focused on *improving theoretical guarantees* of the ASkewSGD algorithm for *quantizing deep neural networks (DNNs)*.
- Eliminated the need for convexity, Lipschitz continuity and differentiability requirements commonly adopted in prior work via Stochastic Approximation.
- Demonstrated ASkewSGD's superiority over benchmarks (BinaryConnect, ProxQuant) in tasks such as synthetic data (TwoMoons) and image classification (MNIST, CIFAR-10) with low-bit quantization for resource-efficient DNN deployment. We publicized reproducible experiments on [GitHub](#).

Frequency Estimation in Strict-Turnstile Data Streams with L_1 α -Bounded Deletions *Spring 2026*
- Extended four *classical streaming algorithms* (Misra-Gries, SpaceSaving, Count-Min Sketch, Count-Sketch) to the strict turnstile model with the L_1 α -bounded deletion property, proving *tighter error bounds that scale with α instead of stream length*.
- Conducted *extensive experiments on Zipfian, uniform, and real-world CAIDA network traces (sliding window and idle timeout streams)*.
- Proposed a novel algorithm, *Dyadic Count-Min Hierarchy with Adaptive Counter Cache*, a hybrid that *combines logarithmic space sampling with a small deterministic cache*, preserving counter-based accuracy for the heaviest items while remaining space efficient for large α .
- Project available on [GitHub](#).

First-Order Methods in Online Learning and Online Convex Optimization *Fall 2024*
- *Studied and implemented key algorithms* such as Online Gradient Descent (OGD), Follow-the-Regularized-Leader (FTRL), and Online Mirror Descent (OMD), analyzing their *regret bounds and convergence behavior under adversarial settings* with strong convexity and smoothness assumptions.
- Conducted empirical evaluations comparing algorithmic performance across synthetic and real-world datasets, focusing on regret minimization and computational efficiency for *online learning / online convex optimization*. The project is available on [GitHub](#).

A Practical Test Set Bound in Binary Classification Theory *Fall 2023*
- Reconstructed and analyzed key proofs by John Langford to deepen understanding of *generalization bounds for binary classification based on merely the empirical risk and size of the training set*, grounded in principles from statistical learning theory.
- Implemented simulations to apply the proposed bounds under given conditions of proposed model performance. The project is available on [GitHub](#).

SELECTED ACADEMIC AWARDS

Best Project Award - UG Summer Research Internship 2025 (Faculty of Engineering, CUHK) *2025*
- *Ranked top 10% among all undergraduate summer research projects*, based on originality, analytical rigor, and professional delivery of the project "Revisiting ASkewSGD: New Theoretical Guarantees for Quantization-Aware Deep Neural Network Optimization". [\[Web Archive\]](#)

Andrew C. Yao Fellowship – HKD \$50,000 *2024*
- Awarded to *only 2 outstanding CSE undergraduates* annually based on *academic distinction and exceptional extracurricular achievements*.
- *Highest distinction in CSE department, named in honor of Professor Andrew Chi-Chih Yao (Turing Award, 2000)*. [\[Hall of Fame\]](#)

ELITE Stream Scholarship – HKD \$20,000

2024, 2026

- Awarded to students of the Engineering Leadership, Innovation, Technology and Entrepreneurship Stream with *outstanding performance in recognized engineering-discipline courses with challenging coursework*. [\[Official Description\]](#)

CUHK Vice-Chancellor's Scholarships for Excellence – HKD \$50,000

2023

- One of the only *8 first-year undergraduate students selected university-wide for exceptional academic and distinguished extracurricular achievements*, recognizing top-tier students with outstanding potential. [\[Official Press Release\]](#)

HKSAR Government Talent Development Scholarship – HKD \$10,000

2023

- For the exceptional achievements in the *innovation, science and technology area (non-academic)*. [\[Official Description\]](#)

Dr. Stanley Ho Medical Development Foundation Scholarship – HKD \$18,000/yr. (renewable)

2022, 2023, 2024, 2025

- Awarded to CUHK students from Macao with *excellent academic performance* (offered by Morningside College, CUHK). [\[College News\]](#)

Dean's List (Faculty of Engineering, CUHK)

2023, 2024, 2025, 2026

- Awarded to Engineering students with *excellent academic performance (Top 10%)*.

COMPETITIONS

Huawei ICT Competition (Hong Kong Regional, Final Round) - Cloud Track

Nov 2022 - Dec 2022

- *Potential Award with HKD \$1,500*, for knowledge and skills related to cloud computing and Huawei's cloud technologies.

International Collegiate Programming Contest (ICPC), Asia Macao Region

May 2021, Apr 2022

- Awarded *Outstanding High School Team* with *MOP \$1,000* (for both years).

Macao Olympiad in Informatics

2018, 2019, 2020, 2021, 2022

- Consistent remarkable performance with an accumulation of *4 Gold Medals (Top 5%)* and *1 Silver Medal* from the first participation onward. [\[Link\]](#)
- Designed, implemented, and tested *efficient algorithms and data structures* in C++ to solve problems under time and memory constraints with *strong logical reasoning and programming proficiency*.
- Competed in *National Olympiad in Informatics (CCF NOI '19)*, *International Olympiad in Informatics (IOI '21, '22)*, *Team of Macao*. [\[Link\]](#)

LEADERSHIP & EXTRACURRICULARS

Vice Chairman and Secretary General - The Macao Association for Competitive Programming

Apr 2026 - Apr 2029 (Est.)

- Government recognized [registered association](#), with the goal of cultivating students' algorithmic thinking through seminars, invited talks, and contests.
- Co-organized the "HKACE 45th Anniversary Cup – HKOI 2026 Asia Invitational" in the Macao Region with Macao Computer Society in May 2026.
- Official Website: [\[Link\]](#); Instagram page: [\[Link\]](#).

Participant - Greece Service-Learning Trip (Organized by Morningside College, CUHK)

June 2024

- Engaged with non-governmental organizations ([Doctors of the World](#), etc.) with the focus on the *Greek NHS and healthcare access for refugees*.
- Performed *critical evaluations* related to *operation structures of NGOs* and how *challenges related to refugees and asylum seekers* are tackled.
- Authored a *comprehensive 10,000-word field journal* and submitted a formal *evaluation report* proposing actionable improvements to healthcare delivery and NGO coordination. Related documents are available [here](#) and [here](#).
- Reference: Morningside College, CUHK ([College News Highlight](#)).



Secretary - The 12th Morningside College Student Union (Executive Council)

Mar 2023 - Feb 2024

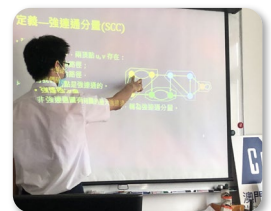
- Maintained accurate meeting minutes and digitalized organizational records, ensuring transparency and continuity across student-led operations.
- Excelled in *drafting activity proposals and managed complex logistics*.
- Led initiatives that fostered *inclusivity and cross-cultural engagement* that enhanced the well-being of Morningside College's diverse and international student community.
- *Organized events and activities* including Graduation Photo Day, Mid-Autumn Festival, Halloween Party, Cultural Fair, Poon-choi Dinner *under strict budget constraints*.
- Reference: Morningside College, CUHK ([Event 1](#) [Event 2](#) [Event 3&4](#) [Event 5](#)).



Member - Macao Computer Society

Sep 2020 - Present

- *Provided comprehensive training programs* online and offline for the Macao team at the International Olympiad in Informatics (IOI) at year 2022, 2023. [\[Selected Session's Slides\]](#)
- *Designed the banner and served as an invigilator for Macao Olympiad in Informatics (MOI) in 2023*, as a member of the organizing committee. [\[Banner\]](#) [\[Ceremony\]](#)
- Served as *invigilator for Macao Olympiad in Artificial Intelligence (MOAI) in 2026*.
- *Proposed 2 original and challenging ICPC-styled competitive programming problems to the Joint-School Invitational Contest* in 2022, designed to test algorithmic thinking and problem-solving skills among secondary school participants, fostering a sense of community and healthy competition.
 - **Problem 1 (Prime Maze)**: Observations in number theory and search algorithms, requiring solvers to identify mathematical patterns to avoid inefficient brute-force solutions. [\[Editorial Available\]](#)
 - **Problem 2 (Matchmaker)**: Basic number theory with (advanced) data structure implementations.
- Reference: Mr. Kuo-Meng SUN (Vice-executive director of MCS).



TECHNICAL SKILLS

Language: Cantonese [Native], Mandarin [Native], English [Proficient] (IELTS Academic 7.5, L-9.0 R-8.5 W-6.5 S-6.5, Test date: Jun 2024).

Programming: C, C++, Java, Python{PyTorch, pandas, openpyxl, scikit-learn, Matplotlib, Django}, OCaml, Prolog, JavaScript, HTML, CSS.

Tools: SQL, LaTeX, Git, WandB, Linux, GEOS-Chem.

IT Certifications: LPI Linux Essentials 010-160 (Score: 800/800); CCF Assessment of Programming Proficiency Level for Secondary Students – Level 7.

Competitive Programming: Efficient and Optimized Computations, Algorithmic Thinking & Problem-Solving, Pedagogical Skills.

COURSEWORKS

Mathematics: Single/Multi-variable Calculus, Linear Algebra, Discrete Mathematics, Probability and Statistics, Game Theory, Ordinary Differential Equations, Optimization Theory, Mathematical Foundations for Data Analytics.

Computer Theory: Data Structures and Algorithms, Formal Languages and Automata Theory, Principles of Programming Languages, Computational Geometry (Postgraduate-Level), Algorithms for Big Data (Postgraduate-Level).

AI&ML: Fundamentals of Artificial Intelligence, Fundamentals of Machine Learning, Numerical Optimization.

Systems: Digital Logic and Systems, Computer Organization, Software Engineering, Operating Systems.

Earth Science: Climate System Dynamics, Solid Earth Dynamics, Ecological Climatology, Cloud Dynamics, Atmospheric Chemistry and Modeling.

Last updated: Aug 31, 2026